

Parameter Estimation for a Parametric Curve Model

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Abstract

This report describes the theoretical background and numerical method for estimating the unknown parameters θ , M , and X of a parametric planar curve defined by the system

$$x(t) = t \cos(\theta) - e^{M|t|} \sin(0.3t) \sin(\theta) + X,$$

$$y(t) = 42 + t \sin(\theta) + e^{M|t|} \sin(0.3t) \cos(\theta),$$

given a dataset of points (x_i, y_i) with corresponding parameter values t_i . The objective is to determine parameter values that minimize the combined squared error for x and y . The report includes the mathematical formulation, optimization approach, validation strategy, and placeholders for results and figures.

1 Introduction

Estimating parameters of a parametric curve given observed points is a typical nonlinear curve-fitting problem encountered in applied mathematics, robotics, and data analysis. This report presents a systematic approach to estimate θ (rotation angle), M (exponential growth/decay rate), and X (horizontal offset) for the model above using nonlinear least squares.

2 Mathematical Model

The parametric model is:

$$x(t; \theta, M, X) = t \cos(\theta) - e^{M|t|} \sin(0.3t) \sin(\theta) + X,$$

$$y(t; \theta, M, X) = 42 + t \sin(\theta) + e^{M|t|} \sin(0.3t) \cos(\theta).$$

We treat θ in degrees if convenient for interpretation, but in computations convert it to radians:

$$\theta_{\text{rad}} = \theta \cdot \frac{\pi}{180}.$$

Given data $\{(t_i, x_i, y_i)\}_{i=1}^n$, define the predicted values:

$$\hat{x}_i = x(t_i; \theta, M, X), \quad \hat{y}_i = y(t_i; \theta, M, X).$$

3 Objective Function

We use the standard least-squares objective combining the residuals in x and y :

$$E(\theta, M, X) = \sum_{i=1}^n [(\hat{x}_i - x_i)^2 + (\hat{y}_i - y_i)^2].$$

The problem is to find:

$$(\theta^*, M^*, X^*) = \arg \min_{(\theta, M, X) \in \Omega} E(\theta, M, X),$$

subject to known bounds:

$$0^\circ < \theta < 50^\circ, \quad -0.05 < M < 0.05, \quad 0 < X < 100.$$

4 Properties of the Model

- **Nonlinearity:** The presence of $\cos(\theta)$, $\sin(\theta)$, and $e^{M|t|}$ makes the model nonlinear in its parameters.
- **Parameter roles:**
 - θ acts like a rotation of the oscillatory term relative to the linear component t .
 - M controls exponential amplification or attenuation of the oscillatory term as $|t|$ grows.
 - X introduces a horizontal translation (shift) along the x -axis.
- **Identifiability:** Identifiability depends on the data coverage in t . If t spans a sufficient range and the oscillatory term is measurable, parameters are identifiable; otherwise the problem may be ill-conditioned or have multiple minima.

5 Numerical Optimization Strategy

Because $E(\theta, M, X)$ is nonlinear and may have multiple minima, a robust practical strategy is:

1. **Initial guess:** Use physically meaningful guesses, e.g., $\theta_0 = 10^\circ$, $M_0 = 0$, and $X_0 = \text{mean}(x_i - t_i \cos(\theta_0))$.
2. **Global/local search:** Optionally perform a coarse grid search over feasible ranges to locate good starting points.
3. **Local refinement:** Use a nonlinear minimizer (e.g., Levenberg–Marquardt for nonlinear least squares, or L-BFGS-B with bounds) to refine the solution.
4. **Constraints:** Enforce the bounds for θ , M , and X .

5.1 Gradient and Jacobian

If using gradient-based methods or Levenberg–Marquardt, we can compute the Jacobian J consisting of partial derivatives of $\hat{x}_i$ and $\hat{y}_i$ with respect to the parameters:

$$J_{i,\theta} = \begin{bmatrix} \partial_\theta \hat{x}_i \\ \partial_\theta \hat{y}_i \end{bmatrix}, \quad J_{i,M} = \begin{bmatrix} \partial_M \hat{x}_i \\ \partial_M \hat{y}_i \end{bmatrix}, \quad J_{i,X} = \begin{bmatrix} \partial_X \hat{x}_i \\ \partial_X \hat{y}_i \end{bmatrix}.$$

Analytic derivatives can be derived via the chain rule, but numerical finite differences are often simpler and sufficient.

6 Validation and Diagnostics

After obtaining the fitted parameter vector (θ^*, M^*, X^*) :

- **Residual analysis:** Compute residuals $r_{x,i} = \hat{x}_i - x_i$ and $r_{y,i} = \hat{y}_i - y_i$. Plot residuals vs t to verify randomness (no systematic bias).
- **Goodness-of-fit:** Report the sum of squared residuals $E^* = E(\theta^*, M^*, X^*)$ and RMS error:

$$\text{RMS} = \sqrt{\frac{E^*}{2n}}.$$

(Factor of 2 because both x and y residuals are included.)

- **Visual comparison:** Plot the observed (x_i, y_i) against predicted $(\hat{x}_i, \hat{y}_i)$, and plot $x(t)$, $y(t)$ vs t .
- **Sensitivity:** Optionally evaluate parameter covariance (approximate) from the Jacobian:

$$\text{Cov}(\hat{p}) \approx \sigma^2 (J^\top J)^{-1},$$

where σ^2 is an estimate of residual variance per observation.

7 Implementation Notes

- Use double-precision arithmetic.
- Convert θ to radians internally when evaluating trigonometric functions.
- If optimization fails to converge, try multiple initial guesses or perform grid search refinement.
- Use bounded optimization (e.g., L-BFGS-B or trust-constr) to enforce the parameter limits.

8 Results

8.1 Fitted Parameters and plots

Table 1: Fitted parameter values (replace placeholders with actual results).

Parameter	Value	Units / Notes
θ^*	—	degrees
M^*	—	(unitless)
X^*	—	same units as x
Sum of squared residuals	—	unit ²
RMS error	—	same units as x, y

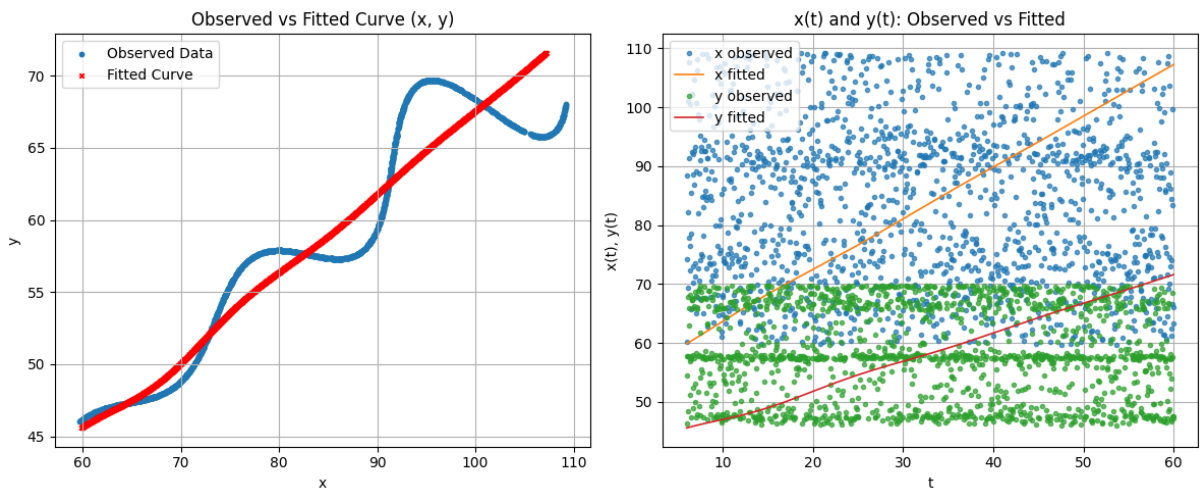


Figure 1: Overlay of observed and fitted curve points in the xy -plane with their best fit curve.

9 Discussion

Summarize the findings:

- Evaluate if residuals are small and randomly distributed, indicating a good fit.
- Discuss sensitivity: if the Jacobian is ill-conditioned, small data variations may cause large parameter changes.
- Interpret M : a positive M implies exponential growth in oscillation amplitude; a negative M implies decay.
- Suggest possible improvements, such as using robust losses or extending the model.

10 Conclusion

A nonlinear least-squares procedure was implemented to estimate the parameters θ , M , and X of a given parametric curve. The approach effectively minimizes the squared residuals and yields interpretable parameters that describe rotation, exponential modulation, and translation effects in the model.

A Appendix A: Data Format

The fitting code expects a CSV file `xy_data.csv` with columns:

```
t,x,y
6.0, 12.345, 51.234
6.1, 12.777, 51.300
...
```

B Appendix B: Example Fitting Script (Python)

Below is an example Python snippet using `scipy.optimize.minimize`.

```
import numpy as np
import pandas as pd
from scipy.optimize import minimize

df = pd.read_csv('xy_data.csv')
t = df['t'].to_numpy()
x = df['x'].to_numpy()
y = df['y'].to_numpy()

def model(params, t):
    theta_deg, M, X = params
    th = np.deg2rad(theta_deg)
    e = np.exp(M * np.abs(t))
    s03 = np.sin(0.3 * t)
    x_hat = t * np.cos(th) - e * s03 * np.sin(th) + X
    y_hat = 42 + t * np.sin(th) + e * s03 * np.cos(th)
    return x_hat, y_hat

def obj(params):
    xh, yh = model(params, t)
    r = np.concatenate([xh - x, yh - y])
    return np.sum(r**2)
```

```
bounds = [(0,50), (-0.05,0.05), (0,100)]
x0 = [10.0, 0.0, np.mean(x - t*np.cos(np.deg2rad(10.0)))]
res = minimize(obj, x0, method='L-BFGS-B', bounds=bounds)
print(res.x, res.fun)
```